

Unless stated otherwise, all problems are worth equal value and answers should be stated as decimals rounded to three decimal places. You must show your work to receive credit.

1. Use figure 0.1 to complete the following.

a. Create a frequency distribution table from figure 0.1 (approximate).

mid	hours	freq.
13	$12 \leq x < 14$	12
15	$14 \leq x < 16$	17
17	$16 \leq x < 18$	13
19	$18 \leq x < 20$	5
21	$20 \leq x < 22$	3
total		50

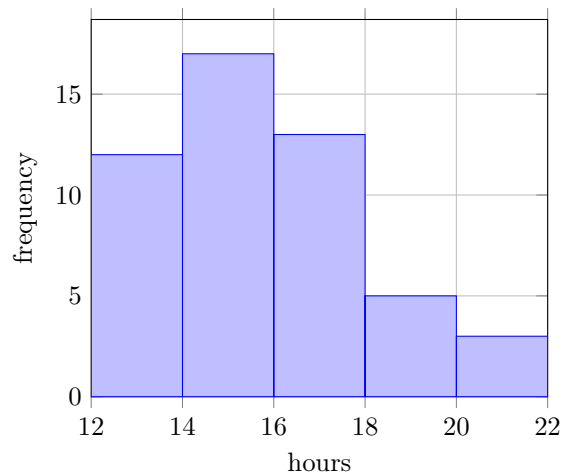


Figure 0.1: Credit hours of 50 students.

b. Using your frequency table, find the mean.

$$\begin{aligned}\bar{x} &= \frac{13 \cdot 12 + 15 \cdot 17 + 17 \cdot 13 + 19 \cdot 5 + 21 \cdot 3}{50} \\ &= \frac{790}{50} = 15.8\end{aligned}$$

c. What percentage of students are taking less than 14 hours?

$$\frac{12}{50} = 24\%$$

2. 10 randomly selected MHU students were surveyed to determine the number of parking tickets they received this semester. The following results were found:

0 7 0 5 1 4 1 2 1 2

a. Find the mean.

$$\bar{x} = \frac{0+7+0+5+1+4+1+2+1+2}{10} = 2.3$$

b. Find the median.

$$\frac{0+2}{2} = 1.5$$

c. Find the mode.

1

d. Find the standard deviation.

$$\begin{aligned}s &= \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} \\ &= \sqrt{\frac{48.1}{9}} \\ &\approx 2.312 \text{ tickets}\end{aligned}$$

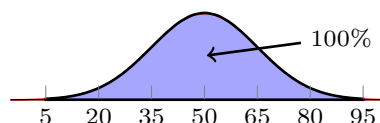
x	(x - $\bar{x}$) ²	Frequency
0	(0 - 2.3) ²	5.29
0	(0 - 2.3) ²	5.29
1	(1 - 2.3) ²	1.69
1	(1 - 2.3) ²	1.69
1	(1 - 2.3) ²	1.69
2	(2 - 2.3) ²	.09
2	(2 - 2.3) ²	.09
4	(4 - 2.3) ²	2.89
5	(5 - 2.3) ²	7.29
7	(7 - 2.3) ²	22.09
total		48.1

3. Two math 107 sections take an exam, and the following is found. The 1st section has a mean score of 75 with standard deviation 15, and the 2nd section has a mean score of 80 with a standard deviation of 10.

Are the scores more consistent in the 1st or 2nd section? Why?

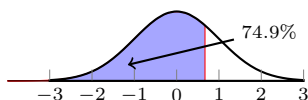
The standard deviation is *smaller* in the 2nd section. So the scores are more consistent in the 2nd section.

4. A population is normally distributed with mean 50 and standard deviation 15. Draw a sketch of the bell curve labeling the mean and the intervals representing 1, 2, and 3 standard deviations of the mean.

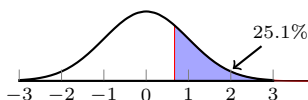


5. Find the following probabilities,

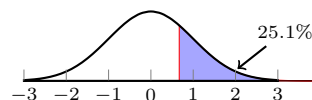
a. $p(z < .67)$
 $\approx 74.86\%$



b. $p(z > .67)$
 $\approx 100\% - 74.86\%$
 $\approx 25.14\%$



c. $p(.09 < z < .67)$
 $\approx 74.86\% - 53.59\%$
 $\approx 21.27\%$

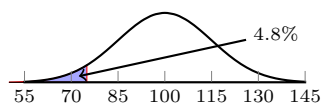


6. A population is normally distributed with a mean 100 and standard deviation 15. Find the following probabilities.

a. $p(x < 75)$

$$z_{75} = \frac{75-100}{15} \approx -1.67$$

$$p(z < -1.67) \approx 4.75\%$$

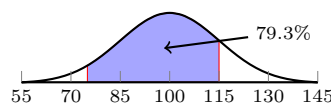


b. $p(75 < x < 115)$

$$z_{115} = \frac{115-100}{15} = 1$$

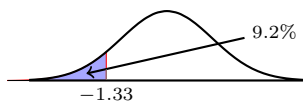
$$p(-1.67 < z < 1)$$

$$\approx 84.13\% - 4.75\% = 79.38\%$$

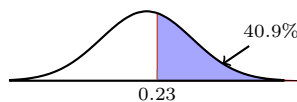


7. Find c such that each of the following is true.

a. $p(z < c) = 9.18\%$
 $.0918 \rightarrow -1.33$
 area $\rightarrow$ z-score
 $c = -1.33$



b. $p(z > c) = 40.95\%$
 $100\% - 40.95\% = 59.05\%$
 $.5905 \rightarrow .23$
 area $\rightarrow$ z-score
 $c = .23$



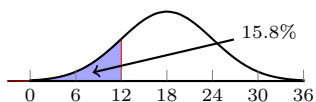
8. [7 points each] The average score on the ACT is 18 with a standard deviation of 6.

a. Find the probability that a random test score is below 12.

Need to find, $p(x < 12)$

$$z_{12} = \frac{12-18}{6} = -1$$

$$p(z < -1) = 15.87\%$$

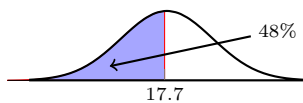


b. What score separates the **bottom** 48.01% from the rest of the population.

Need to find c such that, $p(x < c) = 48.01\%$

$$p(z < -.05) = 48.01\%$$

$$x = (-.05) \cdot 6 + 18 = 17.7$$



9. [BONUS] The *Central Limit Theorem* states that for large enough n , then the means of all samples of size n will be normally distributed. With this in mind, suppose we take a large, randomly selected sample of men and find that $\bar{x} = 70.5$ inches. If it is known that $\sigma = 1.5$ inches, how can we say that this value is within 3 inches of the actual mean height, μ , with 95.4% certainty?

The Central Limit Theorem tells us that the population of sample means will be normally distributed.

So the probability of randomly selecting a sample with 3 inches of μ is:

$$p(-2 \cdot 1.5 < \bar{x} < 2 \cdot 1.5) \approx 95.4\%$$

